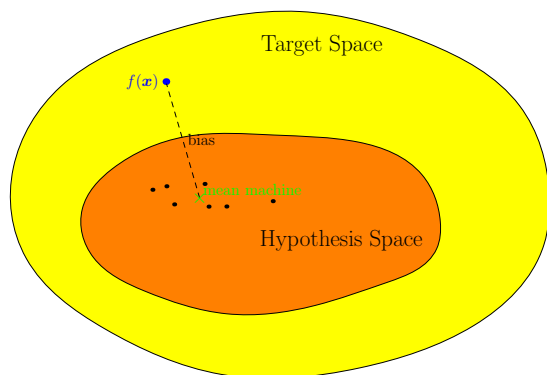
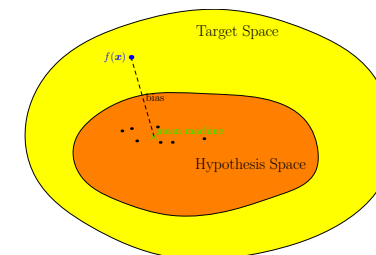


## When Machine Learning Works



When ML Works, Bias Variance

1. What Makes a Good Learning Machine?
2. Bias-Variance Dilemma



## What Makes a Good Learning Machine?

- We want to understand why some machine learning techniques work well and other don't
- To understand why these works we need to understand what makes a good learning machine
- For this we have to get conceptual and think about **generalisation** performance

***generalisation:** how well do we do on unseen data as opposed to the training data*

## What Makes Machine Learning Hard?

- Typically we work in high dimensions (i.e. have many features)
- The problem can be over-constrained (i.e. we have conflicting data to deal with)—solve by minimising an error function
- The problem can be under-constrained (i.e. there are many possible solutions that are consistent with the data)—need to choose a plausible solution
- Typically in machine learning the data will be over-constrained in some dimensions and under-constrained in others
- We can't visualise the data to know what is going on

## Least Squared Errors

- Suppose we want to learn some output  $y$  for a feature vector  $\mathbf{x}$
- We construct a learning machine that makes a prediction  $\hat{f}(\mathbf{x}|\boldsymbol{\theta})$
- We typically choose the machine to minimise a *training loss*

$$L_T(\mathcal{D}) = \sum_{(\mathbf{x}, y) \in \mathcal{D}} \left( \hat{f}(\mathbf{x}|\boldsymbol{\theta}) - y \right)^2 = \sum_{i=1}^m \left( \hat{f}(\mathbf{x}_i|\boldsymbol{\theta}) - y_i \right)^2$$

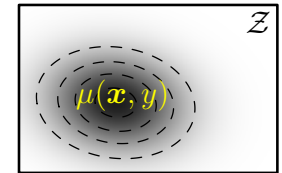
where  $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^m$  is a set of size  $m$ , sampled from a probability distribution  $\mu(\mathbf{x}, y)$

- We call this machine  $\hat{f}(\mathbf{x}|\boldsymbol{\theta}_{\mathcal{D}})$

## Generalisation Error

- We want to minimise the *generalisation loss* which in this case is

$$L_G(\boldsymbol{\theta}_{\mathcal{D}}) = \sum_{(\mathbf{x}, y) \in \mathcal{Z}} \mu(\mathbf{x}, y) \left( \hat{f}(\mathbf{x}|\boldsymbol{\theta}_{\mathcal{D}}) - y \right)^2$$

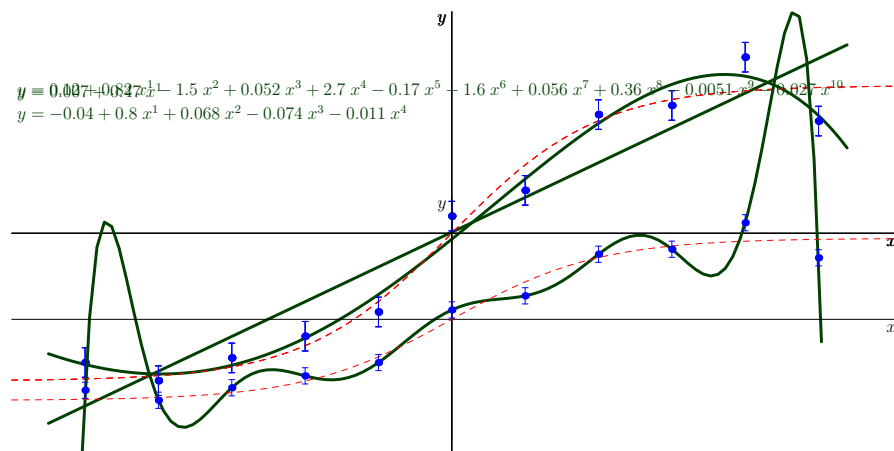


(we can estimate this if we have some labelled examples  $(\mathbf{x}_i, y_i)$  which we have not trained on)

- We want to minimise  $L_G(\boldsymbol{\theta}_{\mathcal{D}})$  but in practice we are minimising  $L_T(\mathcal{D})$  what could possibly go wrong?

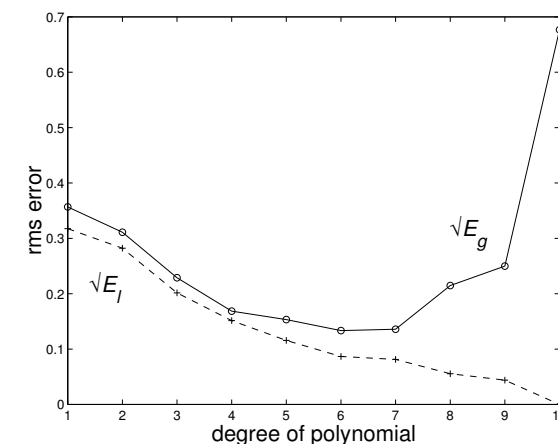
## Too Simple or Too Complex?

- Fit  $\hat{f}(\mathbf{x}|\boldsymbol{\theta}_{\mathcal{D}})$  to data



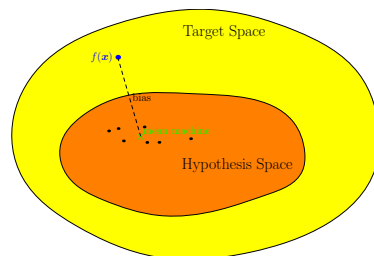
## Measuring Generalisation Error for Regression

- Consider the regression example. The root mean squared error is



## Outline

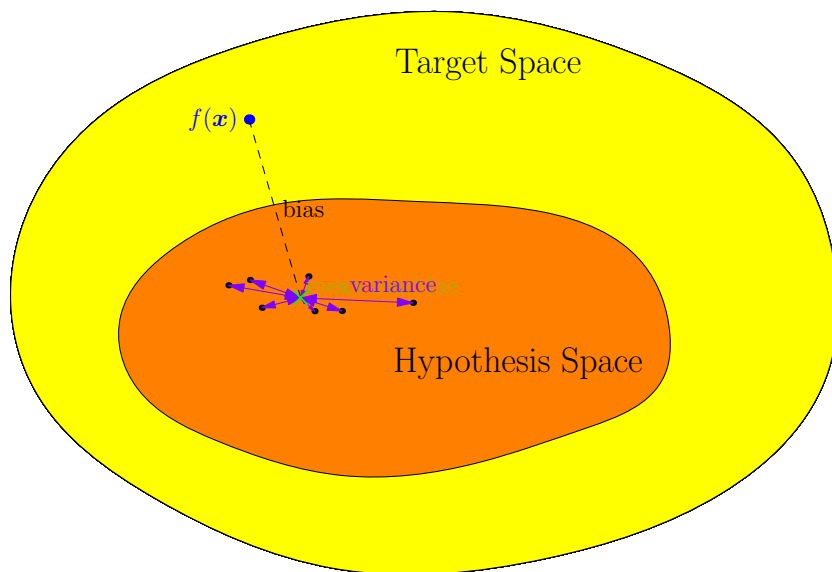
1. What Makes a Good Learning Machine?
2. **Bias-Variance Dilemma**



## Expected Generalisation Performance

- Our generalisation performance will depend on our training set,  $\mathcal{D}$
- To reason about generalisation we can ask what is the *expected generalisation loss*, when we average over all different data sets of size  $m$  drawn independently from  $\mu(x, y)$
- For each data set,  $\mathcal{D}$ , we would learn a different approximator  $\hat{f}(x|\theta_{\mathcal{D}})$
- Note that in practice we only get one data set. We might be lucky and do better than the expected generalisation or we might be unlucky and do worse

## Approximation and Estimation Errors



## Mean Machine

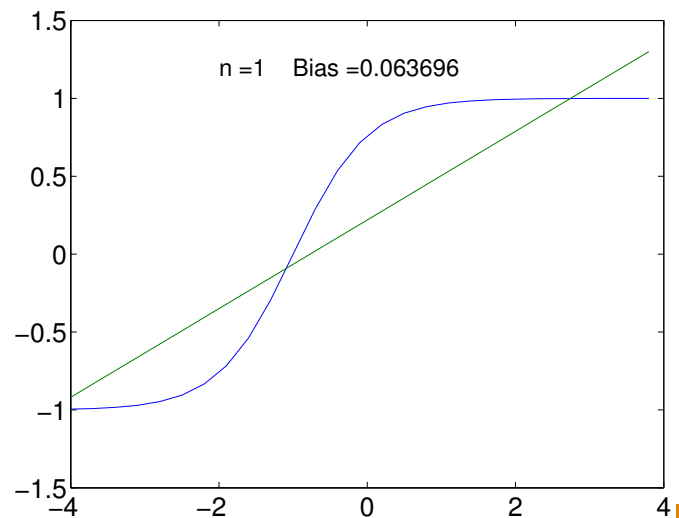
- To help understand generalisation we can consider the mean prediction with respect to machines trained with all data sets of size  $m$

$$\hat{f}_m(x) = \mathbb{E}_{\mathcal{D}} [\hat{f}(x|\theta_{\mathcal{D}})]$$

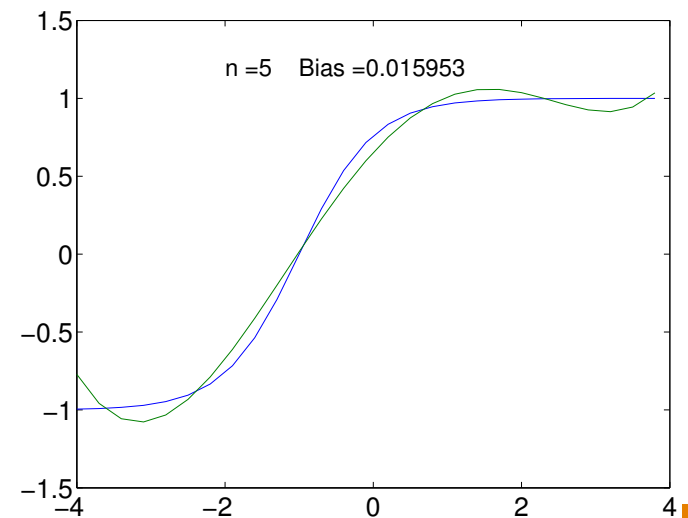
- We can define the **bias** to be generalisation performance of the mean machine

$$B = \sum_{x \in \mathcal{X}} \mu(x, y) (\hat{f}_m(x) - y)^2$$

## Regression Example $n = 1$



## Regression Example $n = 5$



## Bias and Variance

Consider the expected generalisation for data sets of size  $|\mathcal{D}| = m$

$$\begin{aligned}
 \bar{L}_G &= \mathbb{E}_{\mathcal{D}}[L_G(\boldsymbol{\theta}_{\mathcal{D}})] = \mathbb{E}_{\mathcal{D}} \left[ \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \left( \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) - y \right)^2 \right] \\
 &= \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) - y \right)^2 \right] \\
 &= \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right)^2 + \left( \hat{f}_m(\mathbf{x}) - y \right)^2 \right] \\
 &= \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \left( \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right)^2 \right] + \left( \hat{f}_m(\mathbf{x}) - y \right)^2 \right) \\
 &\quad + 2 \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right) \left( \hat{f}_m(\mathbf{x}) - y \right) \right]
 \end{aligned}$$

## Cross Term

- The cross term vanishes

$$\begin{aligned}
 C &= \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right) \left( \hat{f}_m(\mathbf{x}) - y \right) \right] \\
 &= \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right) \right] \left( \hat{f}_m(\mathbf{x}) - y \right) \\
 &= \left( \mathbb{E}_{\mathcal{D}} \left[ \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) \right] - \hat{f}_m(\mathbf{x}) \right) \left( \hat{f}_m(\mathbf{x}) - y \right) \\
 &= \left( \hat{f}_m(\mathbf{x}) - \hat{f}_m(\mathbf{x}) \right) \left( \hat{f}_m(\mathbf{x}) - y \right) = 0
 \end{aligned}$$

- Thus

$$\bar{L}_G = \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right)^2 + \left( \hat{f}_m(\mathbf{x}) - y \right)^2 \right]$$

## Bias and Variance

- We can write the expected generalisation loss as

$$\mathbb{E}_{\mathcal{D}}[L_G(\theta_{\mathcal{D}})] = \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \theta_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right)^2 \right] \\ + \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \left( \hat{f}_m(\mathbf{x}) - y \right)^2 = V + B$$

- Where  $B$  is the bias and  $V$  is the variance defined by

$$V = \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \theta_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right)^2 \right]$$

- Warning some of you will see a similar, but subtly different calculation elsewhere—here  $\bar{L}_G = B + V$

## Bias-Variance Dilemma

- The bias measure the generalisation performance of the *mean machine* and is large if the machine is too simple to capture the changes in the function we want to learn
- The variance measures the variation in the prediction of the machines as we change the data set we train on

$$V = \sum_{\mathbf{x} \in \mathcal{X}} \mu(\mathbf{x}, y) \mathbb{E}_{\mathcal{D}} \left[ \left( \hat{f}(\mathbf{x} | \theta_{\mathcal{D}}) - \hat{f}_m(\mathbf{x}) \right)^2 \right]$$

- The variance is usually large if we have a complex machine
- Striking the right balance is often the key to getting good results

## Balancing Bias and Variance

- We want to choose a learning machine that is complex enough to capture the underlying function we are trying to learn, but otherwise as simple as possible
- There are a number of tricks to achieve this balance
- Some require us to preprocess the data to reduce the number of inputs
- Some machines cleverly adjust their own complexity
- This course looks at machines that achieve this balance

## Lessons

- This course is about understanding machine learning techniques that work well
- Which one to use will depend on the data set
- One of the most useful intuitions about what works is the bias-variance framework
- The bias is high for simple machines that can't capture the data
- The variance is high for complex machines that are sensitive to the training set
- Good machines are powerful enough to capture complex data sets, but they can control their own capacity (ability to (over-)fit the data)